

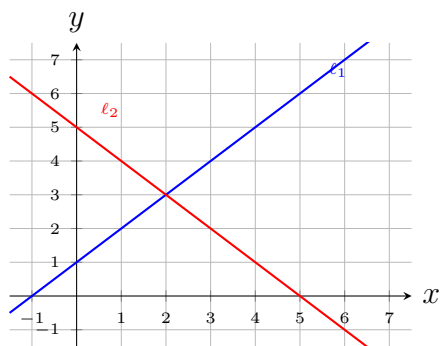
Interpreting the Solution of a Linear System on a Graph

Reading the solution point off a graph, graphing a system to find its solution, and classifying a system as one solution, no solution, or infinitely many

The big idea, and how to use it.

- A point lies on a line exactly when its coordinates make that line's equation true. A point on *both* lines makes *both* equations true — that is what “solution of the system” means.
- To read a solution off a graph, find where the lines cross and write the crossing as an ordered pair (x, y) — across first, then up. Then **check it in both equations**: a graph read by eye can be a square off.
- To solve by graphing, put each equation in slope-intercept form, plot the y -intercept, step off the slope to a second point, draw the line, and read the crossing.
- Two lines can cross once (one solution), never (parallel: no solution), or everywhere (the same line: infinitely many). Equal slopes with different intercepts means parallel; equal slopes *and* equal intercepts means one line drawn twice.

1. The graph shows $\ell_1 : y = x + 1$ and $\ell_2 : y = -x + 5$. Read the solution, then check it in both equations.



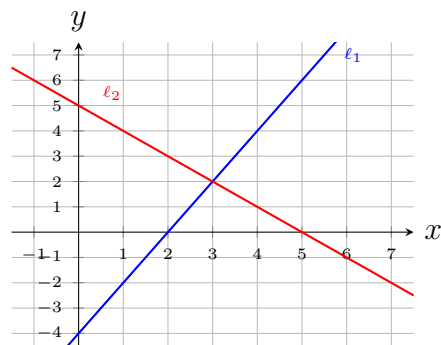
Solution: (_____, _____)

Check in ℓ_1 :

Check in ℓ_2 :

Answer: _____

2. The graph shows $\ell_1 : y = 2x - 4$ and $\ell_2 : y = -x + 5$. Read the solution, then check it in both equations.



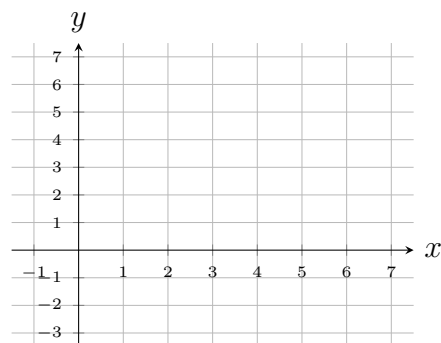
Solution: (_____, _____)

Check in ℓ_1 :

Check in ℓ_2 :

Answer: _____

3. Graph $y = x - 2$ and $y = -2x + 4$ on the grid, then read the solution of the system from your graph.



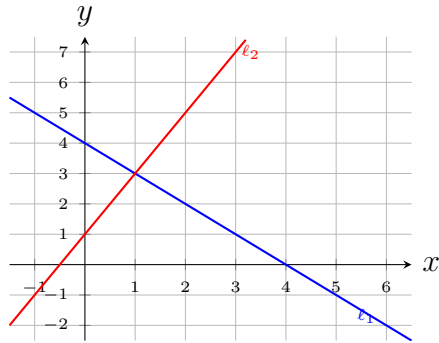
Slope-intercept form and two points for each line:

Solution: (_____, _____)

Check:

Answer: _____

4. The graph shows $\ell_1 : x + y = 4$ and $\ell_2 : y = 2x + 1$. Notice that ℓ_1 is not in slope-intercept form — it does not have to be for you to read the picture. Verify your pair in *both* original equations.



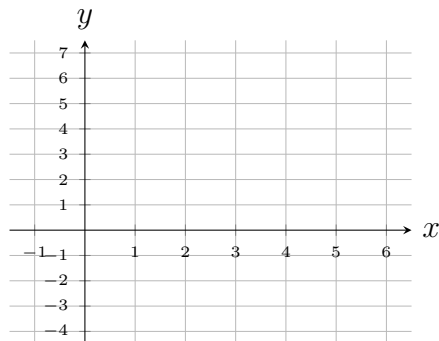
Solution: (_____, _____)

Check in $x + y = 4$:

Check in $y = 2x + 1$:

Answer: _____

5. Graph $y = 2x + 1$ and $y = 2x - 3$. How many points lie on both lines? Say how many solutions the system has and what the graph shows that tells you so.



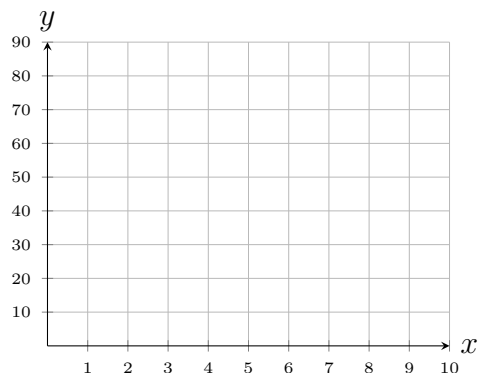
Slope of each line:

Number of solutions:

What the graph shows:

Answer: _____

6. Break-even. Two shops rent kayaks. Let x be the number of hours and y the total cost in dollars. Harbor Rentals charges a launch fee plus an hourly rate: $y = 15 + 5x$. Bayside Boats charges by the hour only: $y = 8x$. Graph both, then say what the crossing point means for a customer.



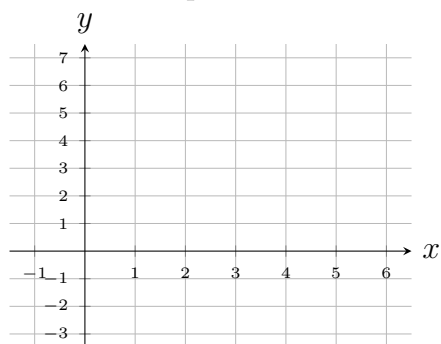
Solution: (_____, _____)

What it means for a customer:

Which shop is cheaper for 3 hours?

Answer: _____

7. Graph $2x + y = 6$ and $4x + 2y = 12$. What do you notice? How many ordered pairs make both equations true? Name two of them, then describe the whole solution set.



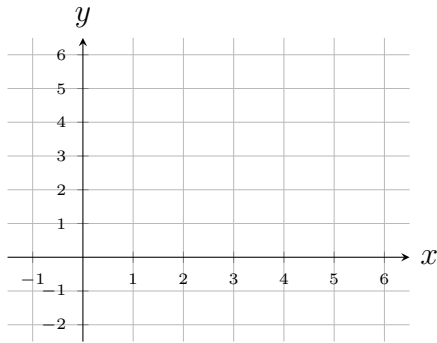
Each equation solved for y :

Two pairs that work:

The whole solution set:

Answer: _____

8. Graph $y = \frac{1}{2}x + 1$ and $y = -x + 4$, read the solution from your graph, and check it in both equations.



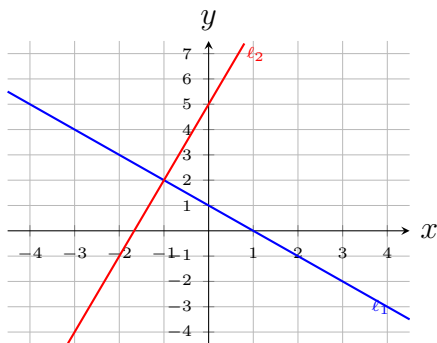
Solution: (_____, _____)

Check in $y = \frac{1}{2}x + 1$:

Check in $y = -x + 4$:

Answer: _____

9. **Find the mistake.** The graph shows $\ell_1 : y = -x + 1$ and $\ell_2 : y = 3x + 5$. A student wrote the solution as $(2, -1)$. Test that point in both equations to show it is wrong, then give the correct solution and say what the student did.



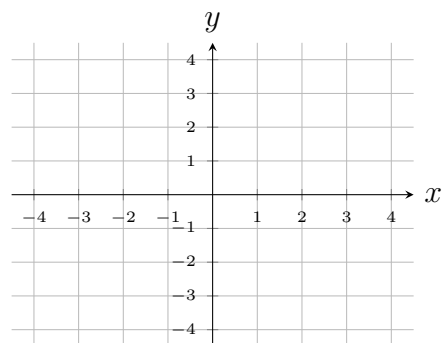
The student's point tested:

Correct solution:

What went wrong:

Answer: _____

10. Explain. Two straight lines are drawn in the same plane. Sketch on the grid to help you think, then explain in two or three sentences why the system cannot have *exactly two* solutions — and why one, none, and infinitely many are the only possibilities.



Your explanation:

Answer: _____